WORKSHEET 2.8

The Other Four Trigonometric Derivatives

Unit 2: The Derivative

Key ideas

- $\dfrac{d}{dx}[\tan x] = \sec^2 x$ and $\dfrac{d}{dx}[\cot x] = -\csc^2 x$.
- $\dfrac{d}{dx}[\sec x] = \sec x \tan x$ and $\dfrac{d}{dx}[\csc x] = -\csc x \cot x$.
- The three **co**-functions (cosine, cotangent, cosecant) have derivatives with a minus sign.
- Each formula follows from the quotient rule applied to sines and cosines.

Worked examples

Example 1 · Simplifying afterwards

Find $\dfrac{d}{dx}[\cot x + x]$.

Solution. $-\csc^2 x + 1$, and $1 - \csc^2 x = -\cot^2 x$.

$\boldsymbol{-\cot^2 x}$**.**

Example 2 · With the product rule

Find $\dfrac{d}{dx}[x \sec x]$.

Solution. $(1)\sec x + x \sec x \tan x$.

$\boldsymbol{\sec x\,(1 + x \tan x)}$**.**

Example 3 · A tangent line

Write an equation of the line tangent to $y = \cot x$ at $x = \dfrac{\pi}{4}$.

Solution. The point is $\left(\dfrac{\pi}{4}, 1\right)$ and the slope is $-\csc^2 \dfrac{\pi}{4} = -2$.

$\boldsymbol{y - 1 = -2\left(x - \dfrac{\pi}{4}\right)}$**.**

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Practice

CORE

1. Find $\frac{d}{dx}[3\tan x]$.

2. Find $\frac{d}{dx}[\csc x]$.

3. For $f(x) = \sec x$, find $f'(0)$.

4. For $f(x) = \cot x$, find $f'\left(\frac{\pi}{2}\right)$.

EXAM LEVEL

5. Find $\frac{d}{dx}\left[\frac{\tan x}{x}\right]$.

6. Find $\frac{d}{dx}[\sec x \tan x]$.

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7. At what value of x in $(0, \pi)$ does the graph of $y = x + \cot x$ have a horizontal tangent?

8. For $f(x) = e^x \tan x$, find $f'(0)$.

Question 9 · Free-response practice

Let $f(x) = 2\sec x - \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

(a) Find $f'(x)$, and show that $f'(x) = \frac{2\sin x - 1}{\cos^2 x}$.

(b) Find the value of x in the interval at which the graph of f has a horizontal tangent.

(c) Find the coordinates of the point on the graph of f at that value of x.

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Worked solutions

1. $3\sec^2 x$.
2. $-\csc x \cot x$.
3. $\sec 0 \tan 0 = 0$.
4. $-\csc^2 \dfrac{\pi}{2} = -1$.
5. $\dfrac{x\sec^2 x - \tan x}{x^2}$.
6. $\sec x \tan x \cdot \tan x + \sec x \cdot \sec^2 x = \sec x \tan^2 x + \sec^3 x$.
7. $1 - \csc^2 x = 0$ gives $\sin^2 x = 1$, so $x = \dfrac{\pi}{2}$.
8. $f'(x) = e^x \tan x + e^x \sec^2 x$, so $f'(0) = 0 + 1 = 1$.
9. (a) $f'(x) = 2\sec x \tan x - \sec^2 x = \dfrac{2\sin x}{\cos^2 x} - \dfrac{1}{\cos^2 x} = \dfrac{2\sin x - 1}{\cos^2 x}$.

   (b) $\sin x = \dfrac{1}{2}$, so $x = \dfrac{\pi}{6}$.

   (c) $f\left(\dfrac{\pi}{6}\right) = 2 \cdot \dfrac{2}{\sqrt{3}} - \dfrac{1}{\sqrt{3}} = \dfrac{3}{\sqrt{3}} = \sqrt{3}$: the point $\left(\dfrac{\pi}{6}, \sqrt{3}\right)$.

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